

Sukhareva Sofia
Hw 1
Bioinformatics Pipelines

Docker

Docker profile run with test data:

```
(base) tori@hamster ~/code/uni/masters/ITMO_BioPipelines_2026 $ ./nextflow run nextflow-io/rnaseq-nf -profile docker

NEXTFLOW ~ version 25.10.4

Launching 'https://github.com/nextflow-io/rnaseq-nf' [prickly_bhabha] DSL2 - revision: ca20a6dfd2 [master]

RNASEQ - NF PIPELINE
=====
transcriptome: /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads         : /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir        : results

executor > local (4)
[0f/5de0bc] process > RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✓
[e8/f33d01] process > RNASEQ:FASTQC (ggal_gut) [100%] 1 of 1 ✓
[39/4d37ef] process > RNASEQ:QUANT (ggal_gut) [100%] 1 of 1 ✓
[e1/5ce301] process > MULTIQC [100%] 1 of 1 ✓

Done! Open the following report in your browser --> results/multiqc_report.html
```

Docker profile run with SRX079566 experiment (human):

```
(base) tori@hamster ~/code/uni/masters/ITMO_BioPipelines_2026 $ ./nextflow run nextflow-io/rnaseq-nf --reads '/home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/SRR{292241,390720}.fastq' --transcriptome /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/Homo_sapiens.GRCh38.cdna.all.fa -profile docker

NEXTFLOW ~ version 25.10.4

Launching 'https://github.com/nextflow-io/rnaseq-nf' [extravagant_celsius] DSL2 - revision: ca20a6dfd2 [master]

RNASEQ - NF PIPELINE
=====
transcriptome: /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/Homo_sapiens.GRCh38.cdna.all.fa
reads         : /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/SRR{292241,390720}.fastq
outdir        : results

executor > local (4)
[23/250f5b] RNASEQ:INDEX (Homo_sapiens) [100%] 1 of 1 ✓
[3d/de440c] RNASEQ:FASTQC (SRR) [100%] 1 of 1 ✓
[3f/834d71] RNASEQ:QUANT (SRR) [100%] 1 of 1 ✓
[13/37ec5e] MULTIQC [100%] 1 of 1 ✓

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 22:25:38
Duration    : 11m 21s
CPU hours   : 0.2
Succeeded   : 4
```

Since the original files were too big to compute in a timely manner, they were cut down to contain only 3000 entries (12k lines). The resulting report will also be in the homework directory.

Conda

Conda profile run with test data:

```
(base) tori@hamster ~/code/uni/masters/ITMO_BioPipelines_2026 $ ./nextflow run nextflow-io/rnaseq-nf -profile conda -resume

NEXTFLOW ~ version 25.10.4

Launching 'https://github.com/nextflow-io/rnaseq-nf' [friendly_solway] DSL2 - revision: ca20a6dfd2 [master]

RNASEQ - NF PIPELINE
=====
transcriptome: /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads         : /home/tori/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir        : results

executor > local (2)
[89/eb1cf3] process > RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1, cached: 1 ✓
[52/1a88ae] process > RNASEQ:FASTQC (ggal_gut) [100%] 1 of 1 ✓
[b5/668f7f] process > RNASEQ:QUANT (ggal_gut) [100%] 1 of 1, cached: 1 ✓
[6b/971cc6] process > MULTIQC [100%] 1 of 1 ✓

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 23:13:00
Duration    : 6m 19s
CPU hours   : (a few seconds)
Succeeded   : 2
Cached      : 2
```